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WHY ECONOMISTS DISAGREE

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(Read November 12, 1964)

IN asking why economists disagree, one evidently takes it as a fact that they do; and, moreover, that they disagree more, in frequency or amplitude, than the learned men professing other disciplines. Evidence of the existence of disagreements among economists is plentiful and notorious. Whether their disagreements are more frequent and more serious than those in other fields is a question to which we shall return later. Certainly their disagreements appear to be greater, and this is enough to justify an inquiry into the causes.

Economists develop general statements on relationships between certain events, magnitudes, and changes of magnitudes, and on the basis of these general statements they offer explanations, predictions, and prescriptions. As long as their discourse is quite abstract, confined to purely deductive inferences from definitions, postulates, and their implications, no one else cares much about whether economists agree or disagree with one another. When they begin, however, to apply their theoretical systems to concrete situations and arrive at conspicuously different explanations, glaringly different predictions, and shockingly different prescriptions, the audience cannot help wondering whether economists really know enough to be taken seriously.

In a comprehensive treatment of my topic I would start with examples of divergent explanations—say, of the Great Depression—then proceed to examples of divergent predictions—say, of post-war economic activity in the United States—and finally come to examples of divergent prescriptions. Such a program would be too big to fit into a twenty-minute nutshell. I shall, therefore, select only one illustration: disagreements on prescriptions for the international monetary system. This choice is indicated because we have recently finished a careful analysis of these disagreements.

DISAGREEMENT ON INTERNATIONAL MONETARY PRESCRIPTIONS

Central bankers had become disgusted with the economists' squabbles and had decided not to

listen to the economists' advice. Understanding this attitude, we proposed an inquiry into the sources of disagreement on international monetary prescriptions. An international study group of 32 economists from eleven countries was brought together for a series of conferences and worked hard on the assigned problems. The group included representatives of several feuding schools of thought. Indeed, quite deliberately, it included extremists on matters of international monetary reform, advocates of the most irreconcilable plans. For only a direct confrontation of the divergent views could afford a full and fair analysis of the sources of disagreement.

The findings of the study were written up and a report was published in August of this year.¹ It will furnish most of the illustrations for my paper today.

THE CHOICE OF POLICIES

To simplify matters I have prepared an exhibit showing some of the differences in the prescriptions of the money doctors concerning the price of gold, the foreign-exchange rates, the foreign reserves held by national monetary authorities, and the targets of domestic monetary policy.

CONFLICTING PRESCRIPTIONS BY THE MONEY DOCTORS

The Price of Gold

1. Leave it unchanged
2. Raise it sharply (double it)
3. Raise it gradually (2 to 4% a year)
4. Reduce it gradually (2 to 4% a year)
5. Reduce it from time to time
6. Leave it to the free market, with governments buying or selling as they like
7. Leave it to the free market, with governments selling their stocks only gradually, avoiding a drastic fall of the price

¹ *International Monetary Arrangements: The Problem of Choice*. A Report on the Deliberations of an International Group of 32 Economists. (Princeton, International Finance Section, Princeton University, 1964.)

Foreign-Exchange Rates

1. Keep them fixed forever
2. Keep them fixed as long as possible, then adjust to new fixed level
3. Keep them pegged, readjust peg from time to time
4. Let them float freely within fixed limits, which are secured by official sales and purchases
5. Let them float without limit but with official sales and purchases at the discretion of the monetary authorities
6. Let them float freely without any official sales and purchases

Foreign Reserves of National Monetary Authorities (Central Banks)

1. Gold, U.S. \$, U.K. £
2. Gold, U.S. \$, U.K. £, French fr., German DM, Dutch fl., etc.
3. Gold, U.S. \$, U.K. £, composite currency unit
4. Gold, composite currency unit
5. Gold only
6. Gold, IMF unit
7. Gold, U.S. \$, U.K. £, IMF unit
8. IMF unit only
9. Nothing

Targets of Domestic Monetary Policy

1. Secure balance of foreign payments at fixed exchange rates
2. Secure stability of domestic price level
3. Secure full employment of labor
4. Secure desired growth rate of gross national product

The list could easily be expanded by adding both to the classes of policy and to the alternative recommendations in each class. But as it stands it will suffice to convince anybody of the fact that the choice of prescriptions is wide and bewildering.

FOUR CAUSES OF DISAGREEMENT

In general we can distinguish four possible causes of disagreement among experts advising on policy: differences as to the meaning of words, differences in logic, differences in factual assumptions, and differences in value judgments.

The first two—semantic and logical differences—can be easily resolved. Differences as to facts can be resolved in principle, if the needed empirical evidence is available; but all too often it is not. Normative differences cannot usually be resolved. But even where facts cannot be ascertained and value judgments cannot be reconciled, the mere identification of the differences can serve an important purpose.

Let us discuss all of these in turn.

DIFFERENT WORD MEANINGS

Terminologies differ and there are no universally accepted definitions of terms. A good deal of economic disagreement is only apparent and disappears as soon as an understanding of word meanings is reached.

Here is a list of words, all used in the discussion of international problems, that mean different things to different economists: money, supply of money, demand for money, velocity of circulation, hoarding, dishoarding, currency, credit, banks, gold standard, devaluation, inflation, deflation, flexible exchange rates, stable price level, full employment, economic growth, balance of payments, financing, surplus, deficit, equilibrium, adjustment, liquidity—and I could go on and on. Yet the situation is not like that in the Tower of Babel, since any group of economists can, if they wish, make an *ad hoc* agreement on the glossary for use in a particular discussion. If, however, one compares published statements by economists not in a semantic cartel, one may get an impression of substantive agreement where the authors actually disagree, or an impression of disagreement where they actually agree with one another.

To illustrate the possibility of semantic confusion in the context of the policy recommendations previously enumerated, let us look at two pertinent terms. "Devaluation" may be understood to refer to (1) an increase in the official price of gold, not necessarily associated with increased prices of foreign currencies; (2) an increase in the official prices of all or some foreign currencies, not necessarily associated with an increase in the price of gold; or (3) increases in both. Arguments for or against "devaluation" will evidently depend on just what is meant. I have avoided this term in my exhibit of conflicting prescriptions.

Similarly, arguments for or against "flexible exchange rates" may depend on what "flexible" is supposed to mean. It may refer to any of the things recommended as points (2), (3), (4), (5), and (6) of our list concerning foreign-exchange rates, and even more variants are possible. Intelligent discussion presupposes agreement on terminology.

DIFFERENT LOGICAL REASONING

We have no real differences in logic—the rules of formal logic are the same for all of us—but there may be logical fallacies behind some of the disagreements. This may happen when two or

more of the factual assumptions on which an argument is supposed to rest are inconsistent with one another—but the inconsistency becomes manifest only after some of the hidden implications of the assumptions are laid bare in an analytic process that calls for even more skill in theorizing than many trained economists can muster.

Here is an example: Some economists contended that a country would develop a deficit in its trade balance if money incomes in foreign countries were increasing at a slower rate than labor productivity, and that the conditions of supply and demand might be such that a devaluation of the currency of the deficit country would not cure the deficit. It was later demonstrated that the conditions under which the deficit would develop and the conditions under which it could not be cured by devaluation are mutually inconsistent.²

It goes without saying that disagreements due to such logical flaws will not persist among competent economists. As soon as the inconsistencies are demonstrated, the argument in question is withdrawn.

DIFFERENT FACTUAL ASSUMPTIONS

The most prolific source of disagreement lies in differences of factual assumptions. It is not customary for experts to state all the assumptions that underlie their conclusions; it would be much too cumbersome. But when they have reached very different conclusions, then we are forced to go back and find out what implicit assumptions they have made. Such assumptions may refer to single events, sequences and conjunctions of events, lasting institutions and temporary constellations of circumstances, economic reactions (mass behavior) and decisions (policies), political reactions and decisions, interrelations between economic and political behavior, etc. Any of these assumptions may be made with respect to the past, the present, or the future.

Assumptions about the past usually relate to matters on which no historical evidence is available. For example, the composition of private short-term capital movements and of "errors and omissions" in the balance of payments of the United States during recent years is unknown and can only be assumed. These assumptions may be pertinent to the solution of the problems with which we are concerned.

² Harry G. Johnson, "Increasing Productivity, Income-Price Trends, and the Trade Balance," *Economic Journal* 64 (Sept., 1954): pp. 462-485.

Assumptions about the present may be similar in nature: they are part of a "diagnosis" based on guesses or estimates not testable at the time, if ever. For example, the amount of the present stocks of gold held by the Soviet Union and of the hoards of gold owned by private investors and speculators are unknown. Lacking this information, we must resort to making assumptions.

Assumptions about the future may be (1) extrapolations and projections of past reactions ("behavior functions"), (2) predictions of events or conditions regarded as probable for a variety of stated or statable reasons, or (3) sheer forecasts based on "insights" of several unspecified kinds. For example, ratios of imports to national income ("propensities to import") are projected into the future; attitudes of central-bank managers which have been typical in the past are supposed to remain unchanged; political group pressures are anticipated to develop under certain conditions and to result in particular decisions by national authorities. The "probabilities" that such assumptions will actually be borne out by the future course of events are debatable. Some of the underlying propositions may become testable in the future, but at present they can, at best, be judged as "sound," "likely," or "realistic" on the basis of verified information which throws some indirect light on them. To bring tacit or implied assumptions of this sort into the open can be essential in an evaluation of differences of opinion.

With respect to economic policy recommendations, it is of particular significance to state clearly all assumptions about political attitudes, because diverse judgments about what is politically "unacceptable," "impossible," or "impractical" may be responsible for wide disagreement among economists. Authorities or legislators charged with making the decisions have a right to know what an economist thinks, not about their likes and dislikes, but about the effects a particular measure, policy, or institution would be likely to have if it were adopted. Hence, the economist should frankly state what he would recommend if the constraint of "political feasibility" were removed, that is, if he could assume the absence of political resistances he thought were preventing the acceptance of certain proposals.

For an illustration of these abstract propositions we may remain close to our previous examples and select from our exhibit the prescription of flexible exchange rates (in one of the forms which minimize official interventions in the market). Sev-

eral economists testified that they proposed other systems chiefly because they assumed that the political resistance to flexible exchange rates was too strong to be overcome. Another group of experts oppose flexible rates and favor fixed rates chiefly because of particular assumptions they make about the attitudes of future managers of central banks; they assume, particularly, that central-bank managers trying to avoid inflationary credit policies (urged upon them by political pressure groups) will succeed more easily when they can point to dwindling foreign reserves under fixed exchange rates than they could by pointing to a declining exchange rate of their currency under a system of flexible rates. (Advocates of flexible rates regard this assumption as an unrealistic speculation about political prejudices prevailing in the future.) Another assumption relevant to the advocacy or rejection of flexible rates relates to the comparative probability that governments resorting to restrictions in the foreign-exchange markets would do so more readily in order to avoid losses of reserves in the case of fixed rates or to avoid depreciation in the case of flexible rates. Again, another assumption relates to the economic effects of possible apprehensions which importers and exporters may have concerning foreign-exchange restrictions, on the one hand, and exchange depreciation, on the other.

Now it should be clear that all these are matters on which we lack knowledge. Nothing else can be done than to make assumptions about what one thinks would be most likely to happen. And nobody can reasonably complain if different experts have different judgments regarding these probabilities.

One more example. The success of a certain monetary arrangement may depend on whether wage rates can be kept from being pushed up faster than productivity increases. Those who assume that no excessive wage-push will occur may prescribe one monetary arrangement, whereas those who fear that the wage-push cannot be avoided may properly prescribe something else. There will be several variants of recommendations according to several possible views about what will happen, may happen, could happen, or might happen. Who can reasonably demand that economists ought to "know" such future developments?

There are some kinds of "behavior functions" about which we may eventually learn more than we know at present and on the stability of which one may possibly be less pessimistic than I seem

to be. Particularly one sect among the economists, the econometricians, have great faith in the stability of numerical relationships in economic life, and they promise us that we shall soon know more about future behavior than we do now. But even they recognize that certain independent variables, or exogenous magnitudes, will always have to be assumed.

Economists need not feel ashamed about their ignorance of an unknowable future. They can do no better than make assumptions, and it would be an inexcusable deception of the public if in the face of extreme uncertainty they conspired to make the same assumptions. Without such collusion, however, it is most unlikely that they will make the same assumptions; consequently, they will arrive at different conclusions and different recommendations.

DIFFERENT VALUE JUDGMENTS

Even identical choices of all relevant assumptions of fact would not secure agreement on policy recommendations if the advisers held different preferences in the pursuit of objectives.³ Indeed, with different aims and purposes, but the same factual assumptions, recommendations cannot possibly be concordant, except in freakish instances of semantic confusion or logical fallacy.

The hierarchy of values ruling our opinions may be quite different from person to person; and even where a broad, general ranking happens to be the same, the comparative weights are likely to differ. The things different people have in mind when they talk about total welfare, and which they group under such headings as total income, employment opportunities, stability, growth, income equality, freedom, etc., cannot all be "maximized" at the same time. As a rule, more of one will mean less of something else; and how much we should be prepared to sacrifice on other scores for the sake of a gain on a particular score is a matter on which people will always differ.

Although we have been told *de gustibus non disputandum*, a discussion on differences in value judgment may be useful. For it happens too often that a partisan of a cause takes a strong position without realizing that only some of his values are "ultimate" (or "terminal") while most are only "instrumental" (or "derived"). An instrumental

³ By "adviser" I mean anybody offering advice, paid or unpaid, upon request or unsolicited, in the government or in the opposition, as an interested party or as a disinterested expert.

value implies a combination of an ultimate value judgment and of factual assumptions about the causal connections between the "instrument" and the presumed effects to which ultimate value is attached. These implied assumptions of fact, however, may be invalid or partially invalid; or the assumptions may be correct in linking the "instrument" with the desired effects, but incomplete in that they disregard some side-effects that would be undesirable even in the advocate's own system of values.

The objectives of a stable price level, full employment, and a faster growth rate are usually lumped together as an economic holy trinity. That they conflict with one another in many circumstances is well known, but that they are also on different levels of intermediacy to ultimate values is commonly overlooked. Fast growth, defined as a sustained rate of increase of income or consumption per head, implies that our children will be much better off than we are and that young people will enjoy increasing incomes as they grow older. This is an ultimate value, or quite close to one. A stable price level, on the other hand, is "good" only in that we may regard it as instrumental to something else. We may assume, for example, that instability in the price level leads to unwanted redistributions of real income. Hence, stability of the price level promotes "equity" in income distribution, which may be an ultimate value for us. (Alas, the actual effects of changing prices upon distribution may be quite uncertain.) Or we may assume that price inflation reduces efficiency in the use of productive resources, which implies smaller outputs of goods and services than might *ceteris paribus* be obtainable at stable prices. Hence, stability of the price level may promote a larger output, which is an ultimate value. (Alas, the assumption that the *cetera* would be *paria* may be questioned.) Or, as a third possibility, we may assume that rising price levels will reduce the rate of saving and the rate of investment, which in turn may reduce the rate of growth. Hence, stability of the price level is valued because it is assumed to promote growth, the ultimate value in this chain of derivation. (Alas, the assumption of this causal connection is sometimes attacked as irrelevant compared with the presumed direct effects upon growth of the expansionary policies which only "incidentally" inflate the price level.)

It happens that my own judgment of the relevant probabilities makes me inclined to accept the

value of price stability. But I recognize that it is not a value *per se* and certainly not an absolute value. If it is instrumental in attaining certain recognized benefits, its realization will usually involve some cost in the form of sacrifices of alternative benefits. This cost is left out of account by single-minded champions of price stability; on the other hand, the cost is probably overestimated by some advocates of the monetary policies which presumably promote full employment and faster growth.

If we understand that many objectives of economic policy are only instrumental values, we come to understand that judgments of fact and judgments of value may be hopelessly tangled. One who favors a certain policy may do so either because of the high value he puts on its expected main effects or because of the low probability he assigns to the occurrence of certain undesired side-effects. For example, those who sing the praises of price stability, and consequently preach austerity and continence in monetary and fiscal policy, usually believe that, as a matter of fact, the employment of labor will not be greatly affected by aggregate spending. Conversely, those who wave the flag of full employment, and consequently cry out for greater courage or indulgence in monetary and fiscal expansion, usually believe that, as a matter of fact, full employment secured by freer spending need not drive up prices. Is it that value judgments affect judgments of fact? Should we perhaps call in one or more psychoanalysts to disentangle the normative and the positive elements in the economists' disagreements?

Some of the values that influence economic policy recommendations are partly instrumental and partly ultimate. For example, the imposition of foreign-exchange restrictions of the discretionary and discriminatory kind (that is, where a government official can grant or reject a request for an allocation of foreign exchange) is opposed by many economists; but some object to such restrictions largely because of probable economic side-effects of the governmental controls—for example reduced efficiency in production—which they regret, while others are chiefly motivated by a distaste for the directly implied reduction of freedom. Both value judgments should first be broken down to bring out what factual assumptions are involved and to what extent ultimate values—national income, personal liberties—are impinged upon.

A similar mixture of values complicates atti-

tudes with regard to "national sovereignty." Certain monetary institutions or arrangements, for example, are being opposed because they are likely to reduce the economic independence, autonomy, or sovereignty of the nation. The judgment in question may be one regarding intermediate values—when it is feared that harmful or unnecessarily costly measures or actions would be imposed upon the nation from abroad; or one regarding an ultimate value—when foreign pressures are resented *per se*, not because of their consequences. In the former case, the ultimate value may be a higher national income, jeopardized by the imposed actions; in the latter case, the ultimate value is the feeling of pride of self-determination. For instance, a nation may be quite willing to take certain actions under anonymous pressures, under the force of circumstances, but resist or reject the same actions if they are recommended by a foreign government or an international body. (In other words, the nation may resent "being told," though it would be prepared to make the same choice of policy by its own free will.) In the first case, the judgment is not about the value of national free will but about other values foregone by the actions recommended or demanded from the outside. A dissection of value judgments of this sort will be required if the sources of disagreement on certain plans for reform of the international monetary system are to be laid bare.

COMPARISONS WITH NATURAL SCIENTISTS

Our exploration of the sources of disagreement on economic policy recommendations has yielded some of the insights required to answer the question raised at the outset: whether disagreements among economists are wider and more numerous than disagreements among practitioners in other fields. Let us make some comparisons with natural scientists, particularly physicists and biologists.

The fact that we have analyzed disagreements on policies is of great importance. For, if there is an impression that natural scientists are broadly in agreement about the greater part of the body of knowledge that constitutes their discipline, this impression is gained from observing them in concerns far removed from policy recommendations. Most physicists or biologists do not feel called upon publicly to offer opinions on public policy involving their fields of study. Those, however, who are engaged in policy development, either in their capacity as government advisers, or as

private citizens speaking out on public affairs, disagree with one another not one bit less than do economists.

Are examples needed to support this point? We need only recall a few more or less recent controversies—about work on the hydrogen bomb, about testing nuclear devices in the air, in water, and underground, about the Mohole project, about the appropriations for research in high-energy physics, about the manned orbital laboratory and the unmanned Mars exploration within the space program—to realize that disagreements among the policy-advising scientists have been ample and fought about with vigor, heat, and often with bitterness. We also find similar entanglement of judgments of value with assumptions of fact. Think only of the question of fallout and its genetic consequences. Or of the question of shelters and the possibilities of reducing nuclear casualties. On issues of this sort one could observe the same phenomenon that we have seen in the economists' controversies: a correlation between normative and positive judgments. Ethical and political inclinations apparently affected the protagonists' views on pure questions of fact, for example, on the effects of radiation through fallout upon mutation of genes.

We have not here discussed why economists disagree on matters of prediction and even explanation, and hence we are less equipped for comparisons with scientists' disagreements on these scores. It is essential, however, to distinguish propositions about sequences of events under specified conditions, assumed as given or actually controlled, from propositions about sequences of events where conditions can be neither controlled nor known with certainty. Impressions of general agreement among scientists refer to the artificial world of the controlled experiment or to a very stable world in which the essential factors are known to remain unchanged (for most practical purposes). When scientists step out of these worlds into a world in which they cannot control all important variables, cannot know their magnitudes, or cannot expect these magnitudes to remain unchanged for sufficiently long periods, disagreements came up thick and fast.

To predict in a world that they do not control, the natural scientists have to make assumptions about unknown or uncertain variables, and their various judgments of relevant facts may differ as widely as the judgments of social scientists differ. The differences and unreliabilities of their predic-

tions are most notorious with regard to such matters as tomorrow's rainstorm, next winter's snow conditions, next autumn's crops, next decade's death rate, or with regard to the exact time it will take for Venice to be sunk or for New York to be drowned under a heightened sea level, or whether we shall find living organisms on Mars, or whether nuclear power will be competitive with conventionally generated energy.

To explain past events in the real-life world, where things happen that have not been expected, is to assume numerous antecedent occurrences and conjoined conditions for which no, or only circumstantial, evidence is available. Thus, the scientists' explanations of particular earthquakes, landslides, floods, conflagrations, explosions, collisions, shipwrecks, and many other past catastrophes may forever remain controversial. In countless litigations expert witnesses testify on opposite sides, supporting their conclusions by different assumptions of fact, not testable with the information available. Similarly, the scientists' explanations of the evolution of the universe, the genesis of stars and planets, the biography of the earth, the origin of life, the evolution of man—all these are based on uncertain knowledge and tentative hypotheses, highly controversial and changing over the years.

The point I am trying to make is this: the exact-construct-world and the controlled-laboratory-world of the scientist are different from the real-life-world.⁴ As long as scientists deal with the

former two worlds, that is, in their theoretical analysis and experimental research, differences of opinion may be infrequent and not conspicuous enough to arouse public notice. As soon as they have to deal with the real-life-world, that is, as forecasters, experts, consultants, strategists, and advisers on public policy, differences of opinion will be wide and frequent. It happens that these are regarded as the nonscientific roles of the scientists, who are best known and respected for their strictly scientific achievements.

Economists also have an exact-construct-world, created for purely theoretical analysis; but no outsider cares what they do with it or appreciates the broad agreement of the analysts about the theoretical system that constitutes their discipline. The economists' work becomes known only where it deals with the real-life-world, in which most things are unknown and almost everything is uncertain. In explaining, predicting, and prescribing for this uncertain world, economists will unavoidably disagree. Some purists among the economists have contended that they, *qua economists*, cannot give advice or make policy recommendations; that those who recommend or prescribe do so in their capacity as citizens or politicians. Such restrictive definitions of economics may be good for their ego but will not help their public image. For, regrettably, it is only in their roles as practical experts, forecasters, consultants, and policy advisers that economists are known to the public. They have thus acquired the reputation of professional wranglers, incapable of ever agreeing with one another.

⁴Fritz Machlup, "Are the Social Sciences Really Inferior?" *Southern Economic Journal* 27 (January, 1961): pp. 173-184.